

Lab Assignment 5: Sub queries and joins in University Database

Objective of this Assignment:

To write SQL query based on the concept of sub query and join.

A) Write the expression for the following set of queries in SQL, based on the set of schema of Assignment (4), using concept of sub query.

1. Find out the names of instructors who work in the same department as the course_id is CS-101.

```
mysql> SELECT NAME
-> FROM INSTRUCTOR
-> WHERE DEPT_NAME = (
->     SELECT DEPT_NAME
->     FROM COURSE
->     WHERE COURSE_ID = 'CS-101'
-> );
```

NAME
Srinivasan
Katz
Brandt

3 rows in set (0.01 sec)

2. Find out the departments whose budget is higher than the budget of 'Biology' department.

```
mysql> SELECT DEPT_NAME
-> FROM DEPARTMENT
-> WHERE BUDGET > (
->     SELECT BUDGET
->     FROM DEPARTMENT
->     WHERE DEPT_NAME = 'Biology'
-> );
```

DEPT_NAME
Comp. Sci.
Finance

2 rows in set (0.01 sec)

3. Find out the students who have taken a course in the same classroom where course_id is CS-101 is held.

```
mysql> SELECT DISTINCT S.NAME
-> FROM STUDENT S
-> JOIN TAKES T ON S.ID = T.S_ID
-> JOIN SECTION SEC ON T.COURSE_ID = SEC.COURSE_ID
-> WHERE SEC.ROOM_NO = (
->     SELECT ROOM_NO
->     FROM SECTION
->     WHERE COURSE_ID = 'CS-101'
->     LIMIT 1
-> );
```

NAME
Zhang
Shankar
Levy
Williams
Chavez

```
5 rows in set (0.02 sec)
```

4. Find out the students who have taken all courses offered by 'Computer Science' department.

```
mysql> SELECT S.ID, S.NAME
-> FROM STUDENT S
-> WHERE NOT EXISTS (
->     (SELECT COURSE_ID
->     FROM COURSE
->     WHERE DEPT_NAME = 'Comp. Sci.')
->     EXCEPT
->     (SELECT COURSE_ID
->     FROM TAKES
->     WHERE S_ID = S.ID)
-> );
```

```
Empty set (0.01 sec)
```

5. Find courses that are taught in a classroom located in the same building as room number 514.

```
mysql> SELECT DISTINCT COURSE_ID
-> FROM SECTION
-> WHERE BUILDING = (
->     SELECT BUILDING
->     FROM CLASSROOM
->     WHERE ROOM_NO = 514
-> );
```

COURSE_ID
BIO-101
BIO-301
HIS-351

```
3 rows in set (0.00 sec)
```

6. Find out the name of the instructor who has not taken semester as spring.

```
mysql> SELECT NAME
-> FROM INSTRUCTOR I
-> WHERE NOT EXISTS (
->     SELECT *
->     FROM TEACHES T
->     WHERE T.ID = I.ID AND SEMESTER = 'Spring'
-> );
+-----+
| NAME   |
+-----+
| Einstein |
| Gold   |
| Califieri |
| Singh  |
| Crick  |
+-----+
5 rows in set (0.00 sec)
```

7. Find out the students who belongs to the department with the highest budget.

```
mysql> SELECT NAME
-> FROM STUDENT
-> WHERE DEPT_NAME = (
->     SELECT DEPT_NAME
->     FROM DEPARTMENT
->     WHERE BUDGET = (
->         SELECT MAX(BUDGET) FROM DEPARTMENT
->     )
-> );
+-----+
| NAME   |
+-----+
| Chavez |
+-----+
1 row in set (0.01 sec)
```

8. Find out the ID and name if the instructor who is the advisor of s_id is 44553.

```
mysql> SELECT I.ID, I.NAME
-> FROM INSTRUCTOR I
-> JOIN ADVISOR A ON I.ID = A.I_ID
-> WHERE A.S_ID = '44553';
+-----+-----+
| ID    | NAME    |
+-----+-----+
| 22222 | Einstein |
+-----+-----+
1 row in set (0.01 sec)
```

9. Find course_id and year for courses taken by “Srinivasan” in the fall semester.

```
mysql> SELECT T.COURSE_ID, T.YEAR
-> FROM TAKES T
-> JOIN STUDENT S ON T.S_ID = S.ID
-> WHERE S.NAME = 'Srinivasan'
-> AND T.SEMESTER = 'Fall';
Empty set (0.00 sec)
```

10. Create a table named SEMESTER_TYPE from SECTION table with two columns named as SEMESTER and YEAR without taking any records from SECTION table.

```
mysql> CREATE TABLE SEMESTER_TYPE AS
-> SELECT SEMESTER, YEAR
-> FROM SECTION
-> WHERE 1=0;
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

11. Insert the semester and year from SECTION table into the SEMESTER_TYPE table who's ROOM_NUMBER is less than 500.

```
mysql> INSERT INTO SEMESTER_TYPE (SEMESTER, YEAR)
-> SELECT SEMESTER, YEAR
-> FROM SECTION
-> WHERE ROOM_NO < 500;
Query OK, 7 rows affected (0.01 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

12. UPDTAE the semester to summer in SEMESTER_TYPE table for the course with course_id equal to CS-315.

```
mysql> UPDATE SEMESTER_TYPE
-> SET SEMESTER = 'Summer';
Query OK, 7 rows affected (0.01 sec)
Rows matched: 7 Changed: 7 Warnings: 0
```

13. Delete from SEMESTER_TYPE table the details of course whose room_number is greater than 1 0 1 .

```
mysql> DELETE FROM SEMESTER_TYPE;  
Query OK, 7 rows affected (0.01 sec)
```

14. Find out the ID, name that has greater tot_cred than some of the students belong to the Comp.sci. department. (Use >some clause)

```
mysql> SELECT ID, NAME  
-> FROM STUDENT  
-> WHERE TOT_CRED > SOME (  
->   SELECT TOT_CRED  
->   FROM STUDENT  
->   WHERE DEPT_NAME = 'Comp. Sci.'  
-> );
```

```
+-----+-----+  
| ID    | NAME    |  
+-----+-----+  
| 00128 | Zhang   |  
| 19991 | Brandt  |  
| 23121 | Chavez  |  
| 44553 | Peltier |  
| 45678 | Levy    |  
| 54321 | Williams|  
| 55739 | Sanchez |  
| 76543 | Brown   |  
| 76653 | Aoi     |  
| 98765 | Bourikas|  
| 98988 | Tanaka  |  
+-----+-----+  
11 rows in set (0.00 sec)
```

15. Find out the name that has greater tot_cred than all the students of Comp.sci. department. (Use >all clause)

```
mysql> SELECT NAME  
-> FROM STUDENT  
-> WHERE TOT_CRED > ALL (  
->   SELECT TOT_CRED  
->   FROM STUDENT  
->   WHERE DEPT_NAME = 'Comp. Sci.'  
-> );
```

```
+-----+  
| NAME    |  
+-----+  
| Chavez  |  
| Tanaka  |  
+-----+  
2 rows in set (0.00 sec)
```

16. Display the details of the instructor where some classes are taken. (Use exist clause)

```
mysql> SELECT *
-> FROM INSTRUCTOR I
-> WHERE EXISTS (
->     SELECT *
->     FROM TEACHES T
->     WHERE T.ID = I.ID
-> );
```

ID	NAME	DEPT_NAME	SALARY
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
45565	Katz	Comp. Sci.	75000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

9 rows in set (0.00 sec)

17. Display the details of the instructor for which no classes are taken. (Use not exist clause)

```
mysql> SELECT *
-> FROM INSTRUCTOR I
-> WHERE NOT EXISTS (
->     SELECT *
->     FROM TEACHES T
->     WHERE T.ID = I.ID
-> );
```

ID	NAME	DEPT_NAME	SALARY
33456	Gold	Physics	87000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00

3 rows in set (0.00 sec)

18. Increase all instructors' salary with salary over 80000 by 8%, and all other instructors receive 6%. (Use case statement)

```
mysql> UPDATE INSTRUCTOR
-> SET SALARY =
-> CASE
->     WHEN SALARY > 80000 THEN SALARY * 1.08
->     ELSE SALARY * 1.06
-> END;
Query OK, 12 rows affected (0.01 sec)
Rows matched: 12  Changed: 12  Warnings: 0
```

B) Write the expression for the following set of queries in SQL, based on the set of schema of Assignment (4), using concept of join.

1. Find out the name of the instructors who teaches any course in summer semester.

```
mysql> SELECT DISTINCT I.NAME
-> FROM INSTRUCTOR I
-> JOIN TEACHES T ON I.ID = T.ID
-> WHERE T.SEMESTER = 'Summer';
+-----+
| NAME |
+-----+
| Crick |
+-----+
1 row in set (0.00 sec)
```

2. Find students who are enrolled in courses offered by the 'Physics' department.

```
mysql> SELECT DISTINCT S.NAME
-> FROM STUDENT S
-> JOIN TAKES T ON S.ID = T.S_ID
-> JOIN COURSE C ON T.COURSE_ID = C.COURSE_ID
-> WHERE C.DEPT_NAME = 'Physics';
+-----+
| NAME |
+-----+
| Peltier |
+-----+
1 row in set (0.00 sec)
```

3. Find the building of the section in any course offered by BIOLOGY department.

```
mysql> SELECT DISTINCT SEC.BUILDING
-> FROM SECTION SEC
-> JOIN COURSE C ON SEC.COURSE_ID = C.COURSE_ID
-> WHERE C.DEPT_NAME = 'Biology';
```

BUILDING
Painter

```
1 row in set (0.00 sec)
```

4. Find out the name and dept_name of student who taught by the instructor whose salary is more than 80000.

```
mysql> SELECT DISTINCT S.NAME, S.DEPT_NAME
-> FROM STUDENT S
-> JOIN TAKES T ON S.ID = T.S_ID
-> JOIN TEACHES TE ON T.COURSE_ID = TE.COURSE_ID
-> JOIN INSTRUCTOR I ON TE.ID = I.ID
-> WHERE I.SALARY > 80000;
```

NAME	DEPT_NAME
Chavez	Finance
Peltier	Physics
Shankar	Comp. Sci.
Williams	Comp. Sci.
Levy	Physics

```
5 rows in set (0.00 sec)
```

5. Find out the building and capacity of the classroom in the department of Comp.sci.

```
mysql> SELECT DISTINCT CL.BUILDING, CL.CAPACITY
-> FROM CLASSROOM CL
-> JOIN SECTION SEC ON CL.ROOM_NO = SEC.ROOM_NO AND CL.BUILDING = SEC.BUILDING
-> JOIN COURSE C ON SEC.COURSE_ID = C.COURSE_ID
-> WHERE C.DEPT_NAME = 'Comp. Sci.';
```

BUILDING	CAPACITY
Packard	500
Taylor	70
Watson	50
Watson	30

```
4 rows in set (0.00 sec)
```

6. Find out the Name of the student whose building is not “Watson”.

```
mysql> SELECT DISTINCT S.NAME
-> FROM STUDENT S
-> JOIN TAKES T ON S.ID = T.S_ID
-> JOIN SECTION SEC ON T.COURSE_ID = SEC.COURSE_ID
-> WHERE SEC.BUILDING <> 'Watson';
+-----+
| NAME   |
+-----+
| Zhang  |
| Shankar|
| Levy   |
| Williams|
| Chavez |
| Brandt |
+-----+
6 rows in set (0.00 sec)
```

7. Find out the Name of the instructors whose classroom capacity is 500.

```
mysql> SELECT DISTINCT I.NAME
-> FROM INSTRUCTOR I
-> JOIN TEACHES T ON I.ID = T.ID
-> JOIN SECTION SEC ON T.COURSE_ID = SEC.COURSE_ID
-> JOIN CLASSROOM C ON SEC.ROOM_NO = C.ROOM_NO
-> WHERE C.CAPACITY = 500;
+-----+
| NAME       |
+-----+
| Srinivasan |
| Katz        |
| Wu         |
| Mozart     |
+-----+
4 rows in set (0.00 sec)
```

8. Find out the semester and grade of the student whose instructor name is Kim.

```
mysql> SELECT T.SEMESTER, T.GRADE
-> FROM TAKES T
-> JOIN TEACHES TE ON T.COURSE_ID = TE.COURSE_ID
-> JOIN INSTRUCTOR I ON TE.ID = I.ID
-> WHERE I.NAME = 'Kim';
Empty set (0.01 sec)
```

9. Find out the name and tot_cred of the students in the fall semester.

```
mysql> SELECT S.NAME, S.TOT_CRED
-> FROM STUDENT S
-> JOIN TAKES T ON S.ID = T.S_ID
-> WHERE T.SEMESTER = 'Fall';
```

```
+-----+-----+
| NAME      | TOT_CRED |
+-----+-----+
| Zhang     | 102     |
| Shankar   | 32      |
| Levy      | 46      |
| Williams  | 54      |
| Zhang     | 102     |
| Shankar   | 32      |
| Peltier   | 56      |
+-----+-----+
7 rows in set (0.01 sec)
```

10. Find all dept_name where the total budget is greater than the average of the total budget at all departments. (Use with clause).

```
mysql> WITH AVG_BUDGET AS (
->     SELECT AVG(BUDGET) AS AVG_VAL
->     FROM DEPARTMENT
-> )
-> SELECT DEPT_NAME
-> FROM DEPARTMENT, AVG_BUDGET
-> WHERE BUDGET > AVG_VAL;
```

```
+-----+
| DEPT_NAME |
+-----+
| Biology   |
| Comp. Sci. |
| Finance   |
+-----+
3 rows in set (0.01 sec)
```